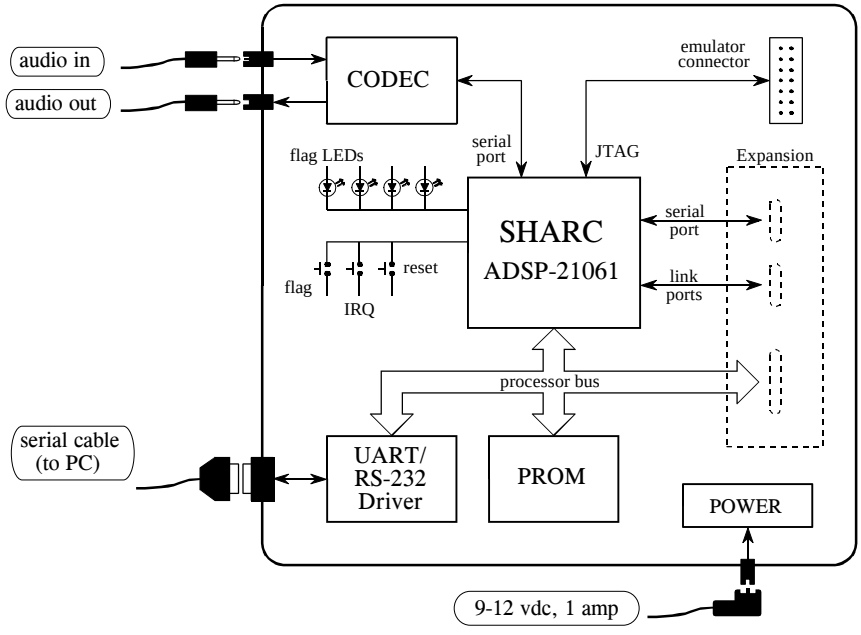

FIGURE 29-1
Block diagram of the EZ-KIT Lite board. Only four external connections are needed: audio in, audio out, a serial (RS-232) cable to your personal computer, and power. The serial cable and power supply are provided with the EZ-KIT Lite.

instance, the ADSP-21060L operates from 3.3 volts, while the ADSP-21160M uses only 2.5 volts.

In June 1998, Analog Devices unveiled the second generation of its SHARC architecture, with the announcement of the ADSP-21160. This features a Single Instruction Multiple Data (SIMD, or "sim-dee") core architecture operating at 100 MHz, an accelerated memory bus bandwidth of 1600 megabytes per second, two 64 bit data busses, and four 80-bit accumulators for fixed point calculations. All totaled, the new ADSP-21160M executes a 1024 point FFT in only 46 microseconds. The SIMD DSP contains a second set of computational units (arithmetic and logic unit, barrel shifter, data register file, and multiplier), allowing ADI to maintain backward code compatibility with the ADSP-2106x family, while providing a road-map to up to ten times higher performance.

The SHARC EZ-KIT Lite

The EZ-kit Lite gives you everything you need to learn about the SHARC DSP, including: hardware, software, and reference manuals. Figure 29-1 shows a block diagram of the hardware provided in the EZ-KIT Lite, based around the ADSP-21061 Digital Signal Processor. This comes as a 4½ × 6½ inch printed circuit board, mounted on plastic standoffs to allow it to sit on